

## **EXPERIMENT 1**

### **Design and configure the hub network using cisco packet tracer**

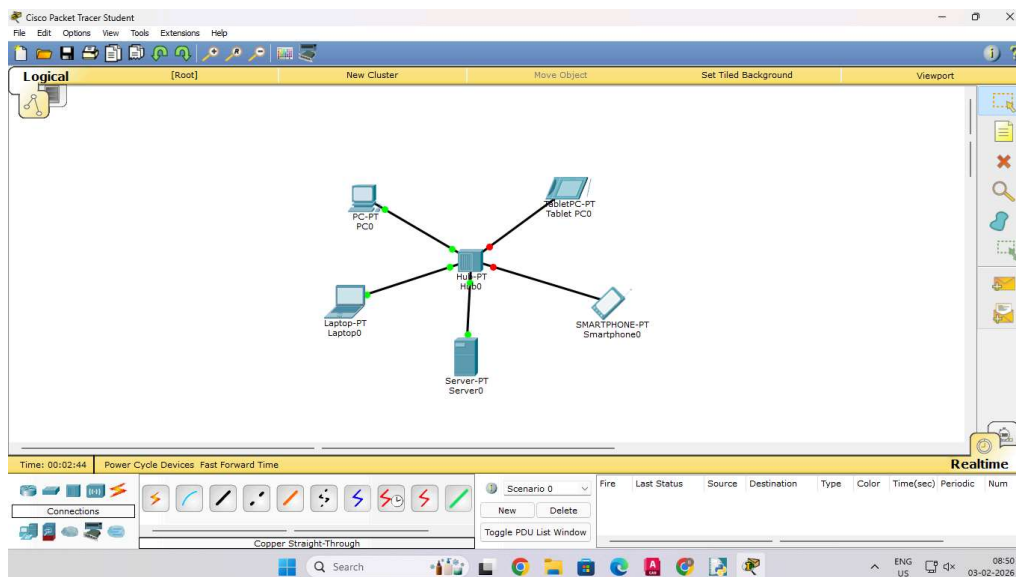
AIM:

To design and configure a simple network using one Hub and five PCs in Cisco Packet Tracer and assign Class A IP addresses to enable communication between the PCs.

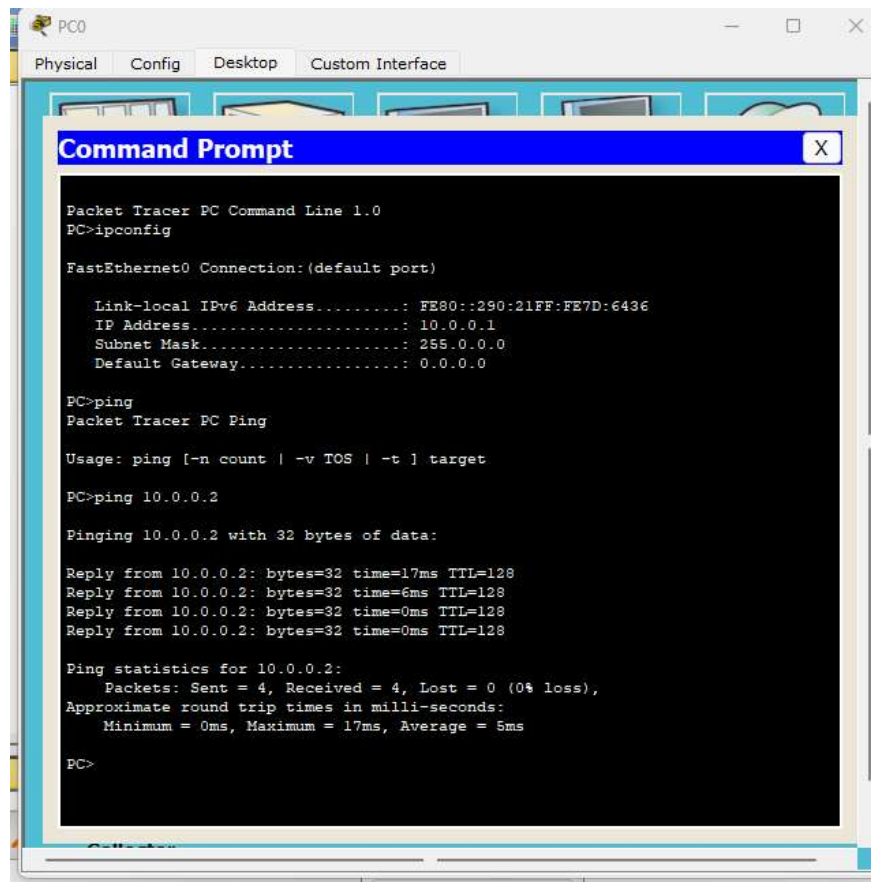
PROCEDURE :

1. Open Cisco Packet Tracer.
2. Drag and place one Hub from the Network Devices section.
3. Drag and place five PCs (PC0–PC4) from End Devices.
4. Connect each PC to the Hub using Copper Straight-Through cables.
5. Assign the following IP addresses to each PC:
  - PC0 → 10.0.0.1
  - PC1 → 10.0.0.2
  - PC2 → 10.0.0.3
  - PC3 → 10.0.0.4
  - PC4 → 10.0.0.5
  - Subnet Mask → 255.0.0.0
6. Verify connectivity using the ping command between PCs

BLOCK DIAGRAM :



OUTPUT :



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PC0
Physical Config Desktop Custom Interface

Command Prompt

Packet Tracer PC Command Line 1.0
PC>ipconfig

FastEthernet0 Connection: (default port)

    Link-local IPv6 Address.....: FE80::290:21FF:FE7D:6436
    IP Address.....: 10.0.0.1
    Subnet Mask.....: 255.0.0.0
    Default Gateway.....: 0.0.0.0

PC>ping
Packet Tracer PC Ping

Usage: ping [-n count | -v TOS | -t ] target

PC>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time=17ms TTL=128
Reply from 10.0.0.2: bytes=32 time=6ms TTL=128
Reply from 10.0.0.2: bytes=32 time=0ms TTL=128
Reply from 10.0.0.2: bytes=32 time=0ms TTL=128

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 5ms

PC>
```

RESULT :

The network was successfully configured, and all PCs communicated with each other using Class A IP addresses.